

Megaloblastic — B12 Metabolism, Absorption & Transport



1. B12 stores 2-3 mg

WHAT YOU SEE

Vault with B12, clock '2-3 mg', '3-4 YEARS'

WHAT IT ENCODES

The liver holds large B12 stores (~2-3 mg) against a daily need of only ~2-5 µg, so deficiency develops VERY slowly — typically 3-4 years after intake/absorption stops. This contrasts sharply with folate (stores last only 3-4 months). Mnemonic: B12 = big vault (years); folate = small fridge (months).

BOARD TRAP

WRONG: expecting a new vegan or post-gastrectomy patient to be deficient within months. Discriminator: pure dietary B12 lack takes years to manifest; rapid B12 deficiency suggests loss of enterohepatic recycling (malabsorption) on top of poor intake, or a coexisting folate problem.

2. Animal sources only

WHAT YOU SEE

'ANIMALS ONLY 5-30 ug/day', meat/eggs/dairy

WHAT IT ENCODES

B12 is synthesized only by microbes and reaches humans through ANIMAL products — meat, fish, eggs, dairy (intake ~5-30 µg/day). Plants contain essentially none, so strict vegans and exclusively breastfed infants of vegan mothers are at risk. Mnemonic: B12 lives in animals, folate lives in foliage (leaves).

BOARD TRAP

WRONG: assuming a balanced vegetarian (eggs + dairy) diet causes B12 deficiency. Discriminator: lacto-ovo vegetarians usually get enough; it is the strict VEGAN with no animal products at all (or impaired absorption) who becomes deficient — and even then slowly because of large stores.

3. Stomach: acid + R-protein

WHAT YOU SEE

Station 1 STOMACH, food protein, acid-pepsin

WHAT IT ENCODES

In the stomach, acid + pepsin cleave B12 from dietary protein; the freed B12 then binds salivary R-protein (haptocorrin/R-binder), which protects it through the acidic gut. Note intrinsic factor is ALSO made here (parietal cells) but does not yet bind B12. Mnemonic: stomach = release + R-binder ride.

BOARD TRAP

WRONG: a patient on long-term PPIs or with atrophic gastritis labeled 'normal absorption.' Discriminator: achlorhydria (PPIs, H2 blockers, atrophic gastritis) impairs the acid-dependent RELEASE of B12 from food, causing food-bound B12 malabsorption — an under-recognized cause of low B12 in the elderly and chronic acid-suppression users.

4. Parietal cells make IF

WHAT YOU SEE

Station 3 'IF BINDING', parietal cells, IF

WHAT IT ENCODES

Gastric PARIETAL cells secrete intrinsic factor (IF), which is mandatory for B12 absorption in the terminal ileum; parietal cells also secrete acid. In pernicious anemia, autoantibodies attack parietal cells and IF, causing both achlorhydria and IF loss. Mnemonic: parietal cells = Pump acid + Provide intrinsic factor.

BOARD TRAP

WRONG (classic): pernicious anemia distractor lists 'anti-smooth-muscle' or 'anti-mitochondrial' antibodies. Discriminator: PA is defined by anti-INTRINSIC-FACTOR (most specific) and anti-PARIETAL-CELL (most sensitive) antibodies, and it is associated with autoimmune gastritis, gastric carcinoid/adenocarcinoma risk, and other autoimmune disease (vitiligo, thyroid).

5. Duodenum: pancreatic enzymes

WHAT YOU SEE

Station 2 DUODENUM, pancreatic trypsin

WHAT IT ENCODES

In the duodenum, pancreatic proteases (trypsin) DEGRADE the R-protein–B12 complex, releasing B12 so it can transfer onto intrinsic factor. Without this step the handoff to IF fails. Mnemonic: pancreas chops the R-binder so IF can grab B12.

BOARD TRAP

WRONG: not connecting chronic pancreatitis / cystic fibrosis / Shwachman-Diamond to B12 deficiency. Discriminator: pancreatic exocrine insufficiency leaves B12 stuck on R-protein (R-protein is not degraded), so it never transfers to IF — a true malabsorptive cause of B12 deficiency that improves with pancreatic enzyme supplementation.

6. Terminal ileum: cubilin

WHAT YOU SEE

Station 5 'TERMINAL ILEUM', cubilin receptor, IF-B12

WHAT IT ENCODES

The IF–B12 complex is absorbed specifically in the TERMINAL ILEUM via the cubilin (cubam) receptor. Therefore terminal-ileal disease (Crohn), ileal resection, bacterial overgrowth (consumes B12), or the fish tapeworm *Diphyllobothrium latum* causes deficiency. Mnemonic: B12's final stop = terminal ileum + cubilin cubbyhole.

BOARD TRAP

WRONG (the board's favorite swap): B12 is absorbed in the 'jejunum.' That is FOLATE. Discriminator: B12 = terminal ileum (needs IF + cubilin); folate = proximal jejunum (PCFT, no IF needed). Crohn ileitis/ileal resection → B12; celiac/jejunal disease → folate.

7. Transcobalamin II delivery

WHAT YOU SEE

Truck 'ACTIVE DELIVERY', 'TC II', 'TC II-B+ MEGALIN'

WHAT IT ENCODES

After absorption, transcobalamin II (TC II) carries the metabolically active fraction of B12 in blood and delivers it to tissues (taken up with megalin). Most circulating B12 rides on haptocorrin (TC I), which is inert — so total serum B12 can mislead. Mnemonic: TC II = the delivery truck that actually drops B12 at the door.

BOARD TRAP

WRONG: trusting a 'normal' total serum B12 to exclude deficiency. Discriminator: most serum B12 is bound to inert haptocorrin; functional deficiency can exist with a borderline-normal level, so confirm with elevated MMA + homocysteine. Holotranscobalamin (active B12) is a more specific marker where available.

8. Enterohepatic recycling

WHAT YOU SEE

Liver 'ENTEROHEPATIC LOOP 0.5-9 ug/day'

WHAT IT ENCODES

B12 is secreted in bile (~0.5-9 µg/day) and largely REABSORBED in the terminal ileum (enterohepatic circulation). This recycling conserves body stores and is the reason dietary lack alone takes years to cause deficiency. Mnemonic: the liver lends B12 to bile and the ileum pays it back.

BOARD TRAP

WRONG: assuming malabsorption and poor diet deplete B12 at the same rate. Discriminator: poor intake still recycles biliary B12 (slow depletion, years), but malabsorption (ileal disease/IF loss) blocks reabsorption of BOTH dietary and biliary B12 — depletion is much faster.

9. Vegans deplete slowly

WHAT YOU SEE

'WHY VEGANS DEplete SLOWLY (recycles biliary B12)'

WHAT IT ENCODES

In pure dietary deficiency (veganism), the absorptive machinery is intact, so biliary B12 is continually reabsorbed in the ileum — this recycling, plus large hepatic stores, means deficiency takes years to appear. Mnemonic: intact ileum = the recycling loop keeps the vegan afloat for years.

BOARD TRAP

WRONG: treating a vegan's slow depletion the same as malabsorptive depletion. Discriminator: the vegan's intact absorption means recycling is preserved (slow); the absorption is the dividing line. Vegans can often be replaced orally, whereas IF-loss/ileal disease classically needs parenteral B12.

10. Malabsorption depletes fast

WHAT YOU SEE

'WHY MALABSORPTION DEPLETES FAST (loses biliary B12)'

WHAT IT ENCODES

Malabsorption (pernicious anemia, ileal resection/Crohn, pancreatic insufficiency, bacterial overgrowth) blocks reabsorption of biliary B12 AND blocks dietary uptake, so the enterohepatic loop is broken — depletion is far faster than pure dietary lack. Mnemonic: a leaky loop drains the vault quickly.

BOARD TRAP

WRONG: giving standard oral B12 to a patient with intrinsic-factor loss/severe ileal disease and expecting correction. Discriminator: IF-dependent absorption is gone, so use parenteral (IM) B12 — or high-dose oral (1000-2000 µg/day) which can be absorbed by IF-independent passive diffusion (~1%).

11. Cobalt corrin ring

WHAT YOU SEE

Cobalt-blue crystals, cobalt atom, corrin ring

WHAT IT ENCODES

B12 (cobalamin) is a large molecule built around a central COBALT atom held in a corrin ring — the cobalt is the catalytic center for both B12-dependent reactions. Mnemonic: COBALamin = COBALt at its core.

BOARD TRAP

WRONG: confusing cobalamin's cobalt-corrin structure with heme's iron-porphyrin ring. Discriminator: B12 = cobalt in a corrin ring; heme = iron (Fe) in a porphyrin ring. Different metal, different ring, different vitamin/cofactor.

12. Methyl- vs adenosyl-B12

WHAT YOU SEE

'ADO-B12', 'METHYL-B12', homocysteine→methionine

WHAT IT ENCODES

B12 has two active coenzyme forms running two reactions: (1) METHYL-B12 is the cofactor for METHIONINE SYNTHASE (homocysteine + methyl-THF → methionine + THF) — links B12 to folate and explains high homocysteine; (2) ADENOSYL-B12 is the cofactor for METHYLMALONYL-CoA MUTASE (methylmalonyl-CoA → succinyl-CoA) — its failure raises methylmalonic acid (MMA). Mnemonic: 'Methyl→Methionine; Adenosyl→methylmalonate (MMA).'

BOARD TRAP

WRONG: thinking elevated MMA occurs in folate deficiency too. Discriminator: the methylmalonyl-CoA mutase (adenosyl-B12) pathway involves NO folate, so MMA rises ONLY in B12 deficiency (normal in folate). Homocysteine rises in BOTH because the methionine-synthase step needs both B12 and folate.